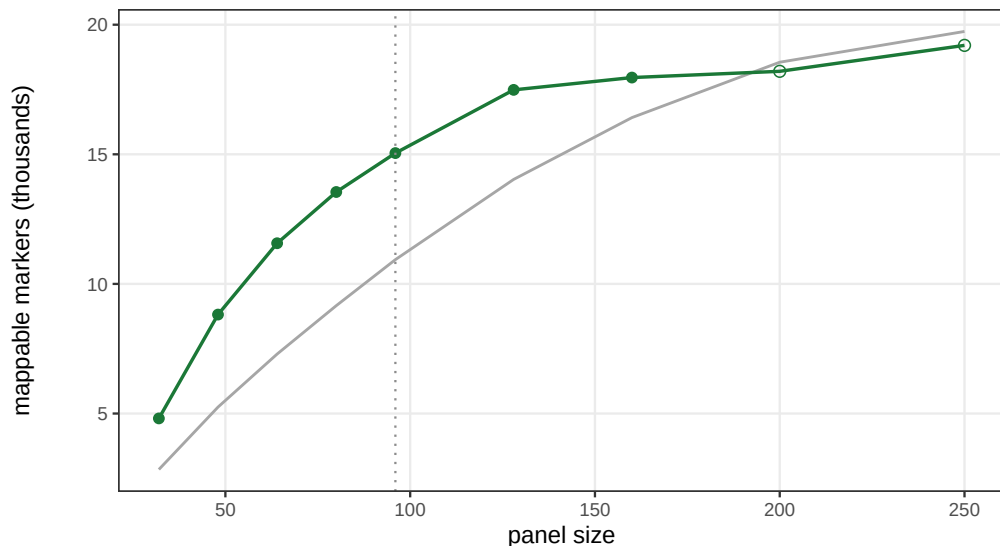


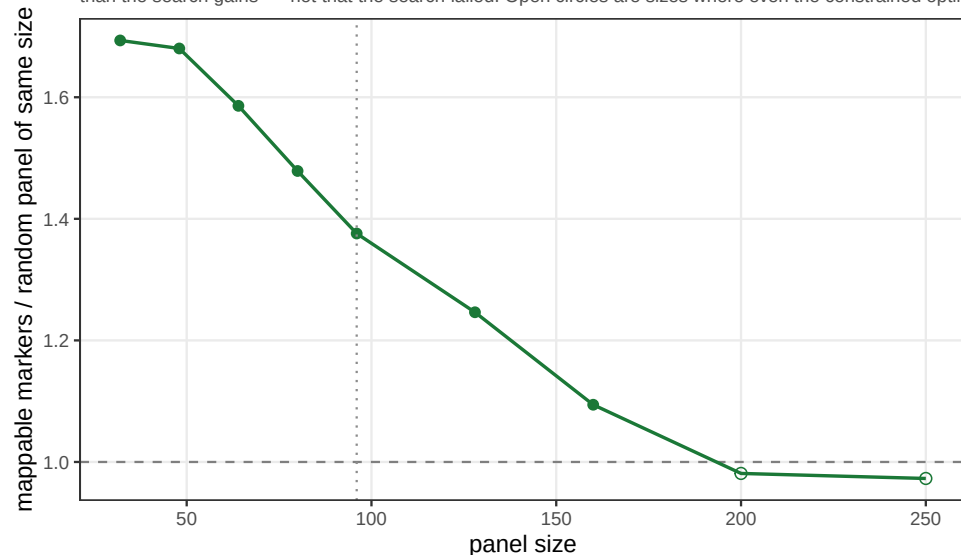
Absolute yield rises with size

Grey is the random-panel mean at each size. On its own this axis just says 'as large as possible'.



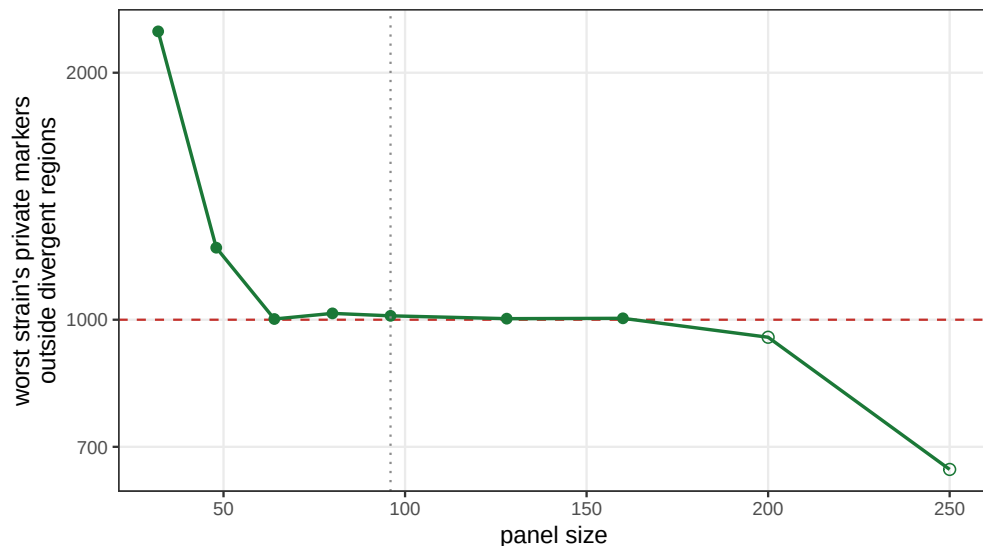
Constrained optimum against UNCONSTRAINED random

Not a like-for-like ratio: almost no random panel meets the identifiability floor --- 2% of 150 draws at $n = 32$ and larger, where the median random panel's worst strain carries 152 private markers at $n = 96$ and 31 at $n = 250$. The null buys its markers with an identifiability it does not have, so a ratio below 1 means the constraint costs more than the search gains --- not that the search failed. Open circles are sizes where even the constrained optimum



The constraint, and where it breaks

Dashed line is $K^* = 1000$. Adding strains destroys privateness, so this can only fall.



Diminishing returns on size

Slope between consecutive sizes. Dotted line marks the 96 used in DIAG_pool_optimizer.R.

